

## MRI (magnetic resonance imaging)

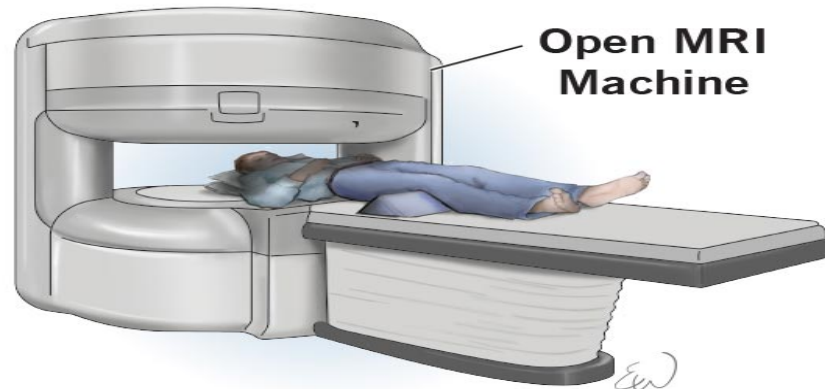
Magnetic resonance imaging, or MRI, is a diagnostic procedure that obtains detailed images of organs and tissues throughout the body, without the need for x-rays or "ionizing" radiation. Instead, MRI uses a powerful magnetic field, radio waves, rapidly changing magnetic fields, and a computer to create images that show disease process, or abnormal condition present.

When the body is placed in a strong magnetic field, such as an MRI scanner, the protons' axes all line up. This uniform alignment creates a magnetic vector oriented along the axis of the MRI scanner. MRI scanners come in different field strengths, usually between 0.5 and 1.5 tesla.

In most medical applications, hydrogen nuclei, which consist solely of a proton, that are in tissues create a signal that is processed to form an image of the body in terms of the density of those nuclei in a specific region. Given that the protons are affected by fields from other atoms to which they are bonded, it is possible to separate responses from hydrogen in specific compounds.

Because MRI doesn't use X-rays or other radiation, it's the imaging test of choice when people will need frequent imaging for diagnosis or treatment monitoring, especially of their brain.

# MRI



There are two main types of MRI machines: closed bore and open. While closed bore MRI machines take the highest quality images, open MRI machines may provide more comfort during the imaging due to the lack of an enclosed space.

What is an open MRI?

Typically, an open MRI machine has two flat magnets positioned over and under patient with a large space between them for patient to lie.

This allows for open space on two sides and alleviates much of the claustrophobia many people experience with closed-bore MRI machines.

However, open MRIs don't take as clear images as closed-bore MRI machines.

Closed-bore MRI machines have a ring of magnets that forms an open hole or tube in the middle where patient lie to get the images. Closed-bore MRIs are narrow with tight head-to-ceiling space. This can cause anxiety and discomfort for some people, but these MRI machines take the best quality images.

What is an MRI with contrast?

Some MRI exams use an injection of contrast material. The contrast agent contains gadolinium, which is a rare earth metal. When this substance is present in the body, it alters the magnetic properties of nearby water molecules, which enhances the quality of the images. This improves the sensitivity and specificity of the diagnostic images.

Contrast material enhances the visibility of the following:

- Tumors.
- Inflammation.
- Infection.
- Blood supply to certain organs.
- Blood vessels.

If MRI requires a contrast material, a healthcare provider will insert an intravenous catheter (IV line) into a vein in hand or arm. They'll use this IV to inject the contrast material.

Some people experience a temporary metallic taste in their mouth after the contrast injection. Contrast materials are safe drugs. Side effects ranging from mild to severe do occur, but severe reactions are very rare.

Prepare for an MRI?

The magnetic resonance imaging (MRI) scanner uses strong magnets and radio wave signals that can cause heating or possible movement of some metal objects in your body. This could result in health and safety issues. It could also cause some implanted electronic medical devices to malfunction.

Leave all jewelry and other accessories or remove them before MRI scan. Metal and electronic items aren't allowed in the exam room because they can interfere with the magnetic field of the MRI unit, cause burns or become harmful projectiles.

Avoid following:

- Heart pacemaker or defibrillator.
- Electronic or implanted stimulators or devices.
- Metallic joint prostheses.
- Cochlear implant or other ear implants.
- Implanted drug pumps, such as those that pump narcotic